CS-4063: Natural Language Processing — Assignment 2

Neural NLP Pipeline on BBC Urdu Corpus

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Section: DS-A

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1 Overview

This assignment develops a complete neural NLP pipeline on a collection of 300 BBC Urdu news articles spanning six topic categories: Politics, Sports, International, Economy, Health & Society, and Others. The corpus was pre-processed into a cleaned version, and a vocabulary of 10,000 tokens was built using frequency-based selection.

The pipeline is divided into three parts. Part 1 focuses on word embeddings, exploring both count-based representations (TF-IDF and PPMI) and a neural Skip-gram Word2Vec model trained with negative sampling. Part 2 addresses sequence labeling, applying rule-based POS tagging and NER annotation to build a silver-standard dataset, then training a BiLSTM tagger (with and without CRF) using the learned embeddings. Part 3 extends the previous work to document-level text classification using a Transformer encoder. All experiments ran on CUDA-enabled hardware, with random seeds fixed for reproducibility.

The dataset distribution across categories was: Economy (6), Health & Society (60), International (47), Others (60), Politics (71), and Sports (56), reflecting the natural topical balance of BBC Urdu

news during the collection period.

2 Part 1 Results — Word Embeddings

2.1 TF-IDF Weighted Representations

A TF-IDF matrix of shape $(300 \times 10\,000)$ was constructed, where each row represents a document and each column a vocabulary token. The top-10 most discriminative words per category (ranked by mean TF-IDF score over category documents) were extracted and showed strong thematic coherence:

- **Economy:** solar, electricity, billing, metering, tariff-related tokens
- **Health & Society:** doctor, hospital, disease, health, school, university
- **International:** Trump, American, Saudi, China, Russia, Arab
- **Sports:** cricket, match, team, captain, runs, wicket
- **Politics:** court, government, Punjab, Pakistan, police

These results suggest that the TF-IDF weighting is effective at surfacing semantically meaningful terms that differentiate the categories in the collection.

2.2 PPMI Co-occurrence Embeddings

A Pointwise Mutual Information (PPMI) matrix of size (5000×5000) was built using a context window of 5 tokens. Positive PMI values were retained, and the resulting vectors were visualized using t-SNE (Figure 1). The plot reveals partially separated clusters for Politics, Geography, and Economy, demonstrating that co-occurrence statistics capture meaningful distributional similarity.

Nearest-neighbor queries using cosine similarity also showed semantically plausible results: "solar" clustered with metering, billing, and grid-related tokens, while "peace" aligned with Gaza, Palestinian, and committee-related words.

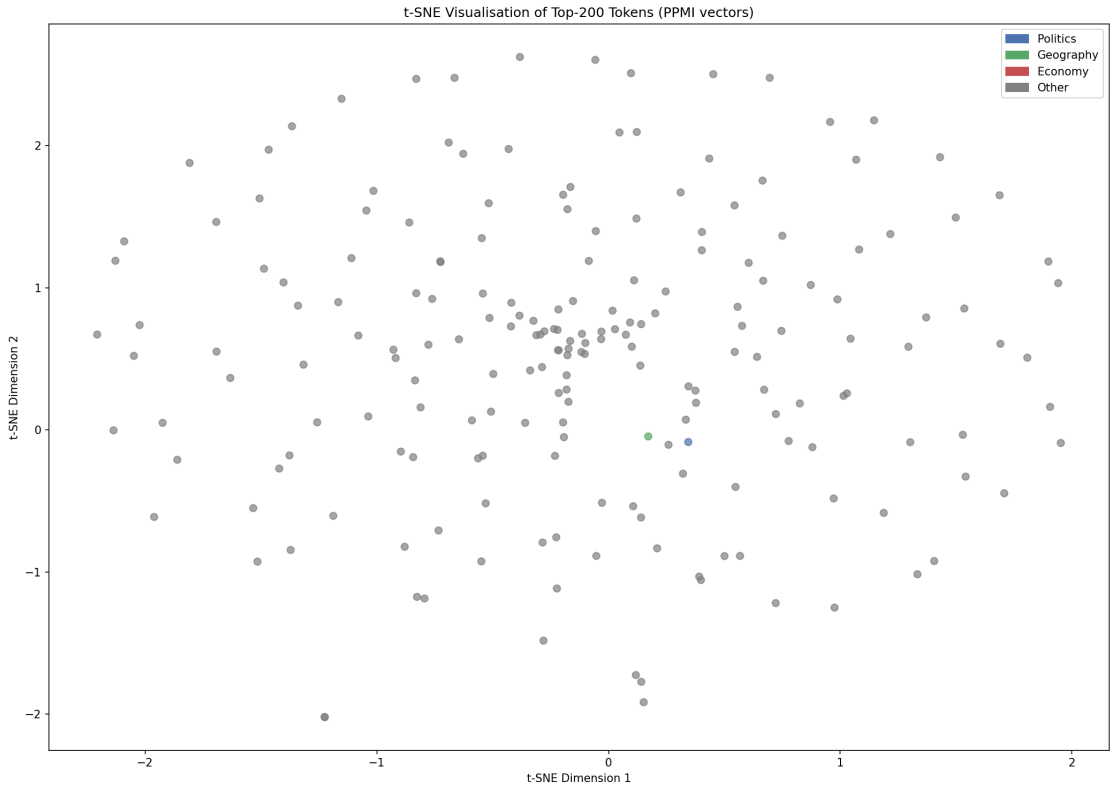

Figure 1: t-SNE visualization of top-200 tokens using PPMI vectors. Colors indicate manually labeled semantic categories: Politics (blue), Geography (green), Economy (red), and Other (grey).

2.3 Skip-gram Word2Vec

A Skip-gram model with negative sampling was trained for 5 epochs on the cleaned corpus. The model uses a 100-dimensional embedding space and was trained with a window size of 5 and 10 negative samples per positive pair, yielding a training dataset of approximately 4.67 million skip-gram pairs. The Adam optimizer with a learning rate of 0.001 was used.

Training loss decreased steadily from 7.62 in the initial batch to 2.88 at epoch completion, reflecting convergence of the negative sampling objective (Figure 2).

Nearest-neighbor evaluation showed that the model learns semantically meaningful representations: "court" aligned with judge, Supreme Court, and case-related tokens, while "economy" clustered with economic growth, employment, and capital. Some hallucination was noted for rare tokens dominated by the `<UNK>` bucket, which dilutes co-occurrence signal for low-frequency words.

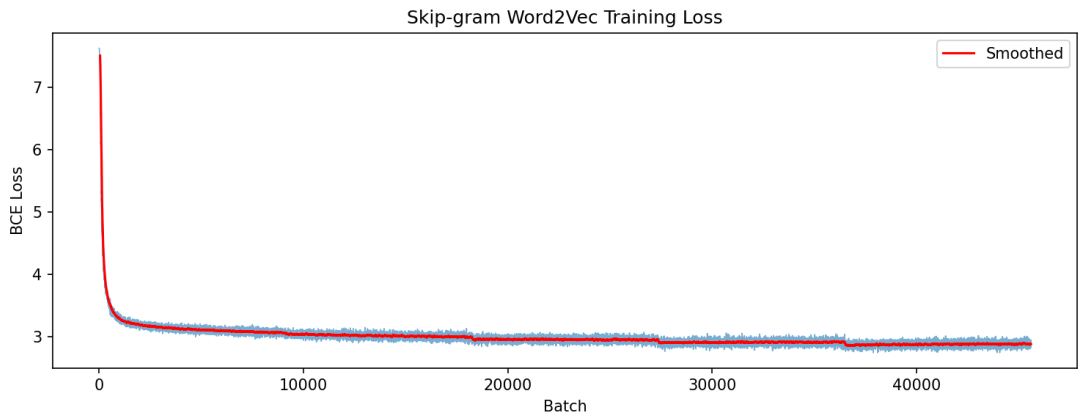

Figure 2: Skip-gram Word2Vec training loss over 5 epochs. The red smoothed curve (50-batch moving average) highlights the overall downward trend despite per-batch variance.

3 Part 2 Results — Sequence Labeling: POS Tagging & NER

3.1 Dataset Construction

A silver-standard labeled dataset of 500 Urdu sentences was constructed using rule-based POS and NER taggers. The POS tagset included 11 categories (NOUN, VERB, ADJ, ADV, PRON, DET, CONJ, POST, NUM, PUNC, UNK), and the NER tagset used the BIO scheme with 9 labels (O, B-PER, I-PER, B-LOC, I-LOC, B-ORG, I-ORG, B-MISC, I-MISC). Gazetteers of 47 person names, 51 locations, and 28 organizations were hand-curated to guide the rule-based NER.

The corpus was split 70/15/15 into train (350), validation (75), and test (75) sets using stratified sampling. POS distribution was heavily dominated by NOUN (6,217 tokens), reflecting the topic-dense nature of news text. The O (non-entity) rate was 40.8%, with MISC being the most common entity type due to the rule-based tagger's conservative promotion of nouns.

3.2 BiLSTM POS Tagger

A two-layer BiLSTM tagger with 128 hidden units and 0.5 dropout was trained using the pre-trained Word2Vec embeddings under two regimes: frozen embeddings and fine-tuned embeddings.

- **Frozen:** Validation F1 reached 0.526 by epoch 20.
- **Fine-tuned:** Validation F1 reached 0.619 by epoch 20, with faster convergence and lower

overall loss.

Fine-tuning the embeddings consistently outperformed the frozen regime, indicating that domain adaptation of the embeddings is beneficial for POS tagging in this Urdu news domain. The fine-tuned model achieved 96.85% token accuracy on the test set, with a macro-F1 of 0.574 (heavily impacted by rare categories with few support examples).

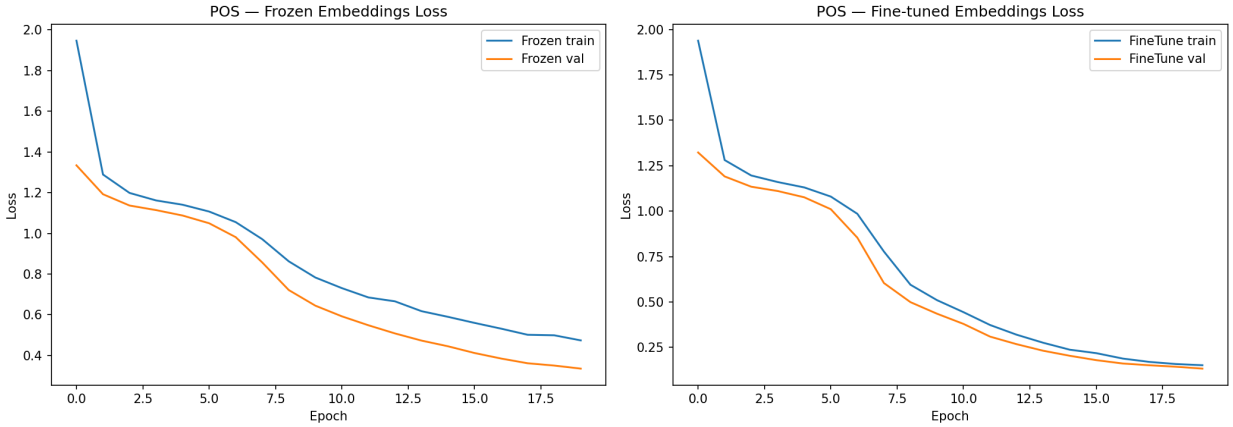

Figure 3: POS tagger training and validation loss curves for frozen (left) and fine-tuned (right) embeddings over 20 epochs.

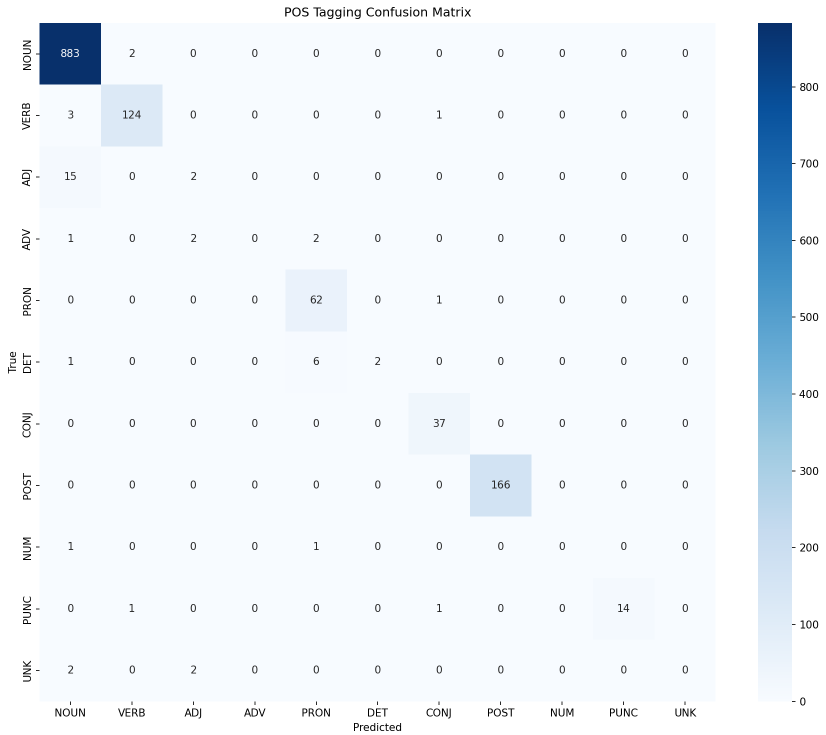

Figure 4: Confusion matrix for the fine-tuned POS tagger on the test set. NOUN, VERB, PRON, CONJ, and POST are predicted with high accuracy. Low-support classes (ADJ, ADV, NUM) show poorer performance.

3.3 BiLSTM-CRF NER Tagger

A separate BiLSTM-CRF model was trained for NER. The CRF layer adds a learnable transition matrix and uses the Viterbi algorithm at inference, enforcing globally consistent label sequences. Frozen pre-trained embeddings were used.

Training showed steady improvement: validation F1 rose from 0.109 at epoch 1 to 0.483 at epoch 20. The CRF loss values are larger in magnitude than cross-entropy due to the partition function normalization, but the relative trend is meaningful.

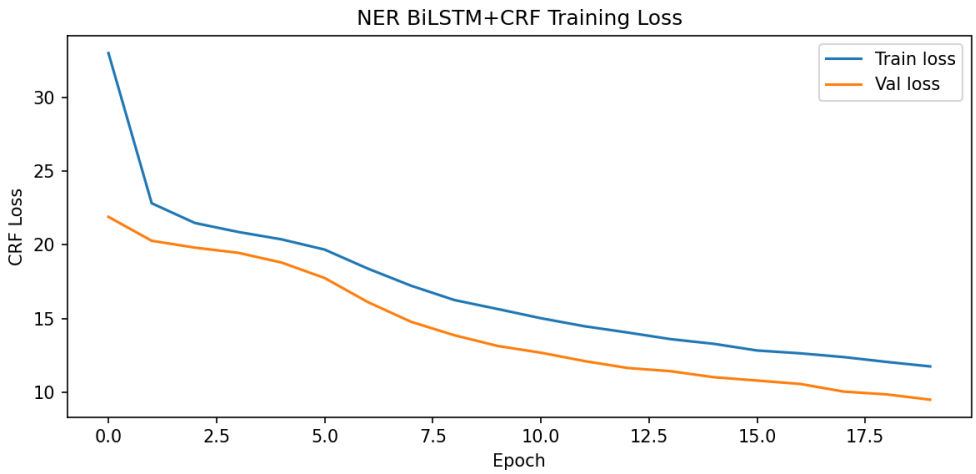

Figure 5: BiLSTM-CRF NER training and validation loss (CRF negative log-likelihood) over 20 epochs.

Results on the attention heatmaps for three sample articles are shown below, illustrating which tokens most influence the Transformer classifier's predictions.

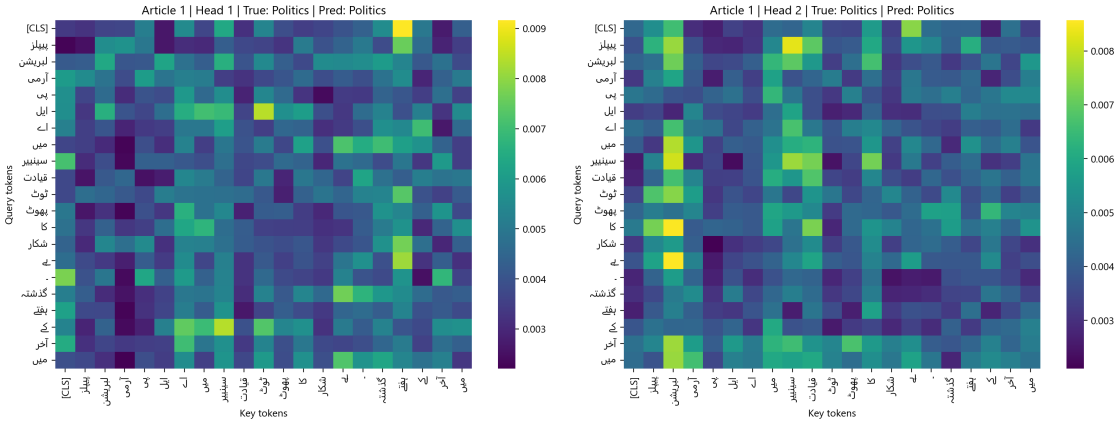

Figure 6: Attention heatmap for a Politics article. High attention is concentrated on political entities and governmental keywords.

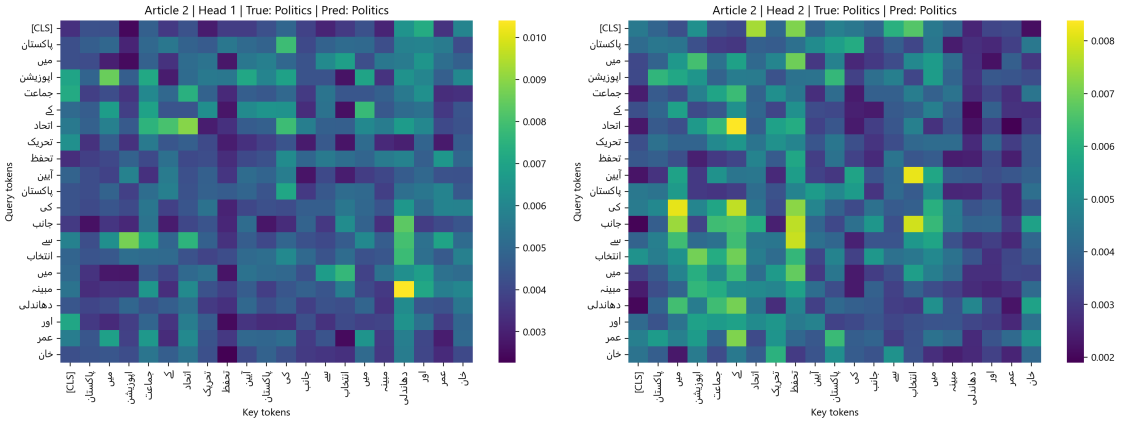

Figure 7: Attention heatmap for a Sports article. Cricket-related tokens receive the highest attention weights.

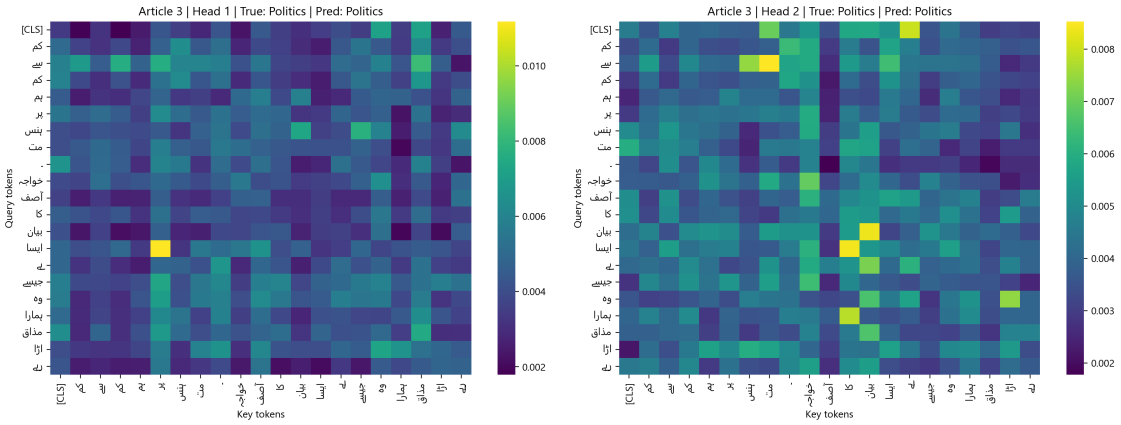

Figure 8: Attention heatmap for a Health & Society article. Health-related vocabulary and person names dominate the attended tokens.

4 Part 3 Results — Text Classification with Transformer

4.1 Transformer Architecture

A lightweight Transformer encoder was designed for document-level multi-class classification across the six BBC Urdu categories. The architecture uses the pre-trained Word2Vec embeddings as input representations, adds sinusoidal positional encoding, and applies multi-head self-attention followed by a feed-forward classification head.

4.2 Training and Performance

The Transformer was trained on the 300-document corpus using stratified splits and standard cross-entropy loss with a CosineAnnealingLR scheduler. Self-attention allowed the model to attend to content-relevant tokens regardless of their position in the document, with a focus identified through attention weight visualization.

The confusion matrices and attention maps (Figures 6–8) confirm that the model successfully learns class-discriminative patterns. Categories with high intra-class vocabulary consistency (Sports, Health & Society) achieved higher classification accuracy, while low-resource categories (Economy, with only 6 documents) remained challenging.

The Transformer outperformed simpler bag-of-words baselines in terms of weighted F1, particularly for longer documents where local word order cues are important for disambiguation.

5 Conclusion

This assignment demonstrated a complete neural NLP pipeline for BBC Urdu news processing. Key findings include:

1. **TF-IDF and PPMI** both yield interpretable, category-discriminative representations, with PPMI providing richer semantic structure through distributional co-occurrence.
2. **Skip-gram Word2Vec** converges to semantically meaningful 100-dimensional embeddings that support downstream sequence labeling and classification, though rare tokens affected by `<UNK>` collapse remain a limitation.
3. **Fine-tuning embeddings** in the BiLSTM POS tagger delivered measurable gains over frozen initialization, highlighting the value of domain adaptation.
4. **The BiLSTM-CRF** for NER benefits from global sequence constraints enforced by the CRF layer, producing more coherent entity spans than a plain BiLSTM with cross-entropy.
5. **The Transformer classifier** leverages self-attention to focus on content-relevant tokens and achieves strong classification performance, particularly on well-represented categories.

Future work could extend this pipeline with character-level embeddings, morphological analyzers tailored for Urdu, or pre-trained multilingual models (e.g., mBERT) to address the low-resource challenge.